

High-Efficiency Microwave Rectifying Circuits with Extended Dynamic Range

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Abstract—In this paper, a microwave rectifier based on elimination of input filter is proposed to operate in a wide range of input power, operating frequency and output load. In the proposed design, a $\lambda_g/12$ short-ended transmission line is inserted between the diode's cathode and ground. This short-ended transmission line presents inductive impedance to compensate the diode capacitive impedance at fundamental frequency and turns to an open circuit at the third harmonic. Thus, both the input low-pass filter and input matching circuit are removed in proposed topology. Theoretical analysis and experiments are both presented. The fabricated rectifier has a very compact dimension of 18 mm by 20 mm. A maximum RF-DC efficiency of 76.6% with 13 dBm input power is achieved. The measured RF-DC efficiency remains above 70% with 5 dB dynamic input power range from 8.25 dBm to 13.25 dBm. At 10 dBm input power, the RF-DC efficiency maintains more than 70% from 5.5 GHz to 6.2 GHz.

Index Terms—Microwave power transmission, Schottky diode, rectifier, harmonic termination, source-pull, high efficiency

I. INTRODUCTION

THE Microwave power transmission system (MPT) is a promising technology and has advantages when the wired power is inconvenient [1]. To improve the overall efficiency of MPT system, the RF-DC (RF energy into DC power) conversion efficiency of rectifier which converts microwave power into direct DC power, should be as high as possible. Other requirements are compact size and high power capacitance, since most microwave rectifiers are install in an antenna array and become rectennas in the microwave power transmission system [2].

With the rapidly development of integrated circuit and semiconductor, MPT technology has been developed to the low-power communication system, such as RFID [3], wireless sensor networks [4] and so on. The developing WPT system has to transfer data and power simultaneously in future. Due to the rapidly increasing data rate and the amount of information, the band of carrier wave is quite wide. Hence, a high power conversion efficiency and wideband rectifier is needed. Currently several broadband and multi-band rectifier are designed [6]–[8]. For instance, a broad-band rectifier is based on maximizing the quality factor of matching circuits [6]. In

paper [7], a broadband rectifier is realized by using branch-line coupler. The modified real frequency technology is applied to optimizing the input matching circuits at input in [8]. However, these broadband matching at the input port definitely raises the insertion loss and occupies more dimensions.

II. PRINCIPLE FOR THE PROPOSED RECTIFIER

A. Principle

A conventional microwave rectifying schematic is shown in Fig. 1, in which a low-pass filter and a DC-pass filter are applied to recycle the harmonics produced during rectification [9]. The harmonics are confined between the input low-pass filter and the output DC-pass filters. The input low-pass filter operates as matching circuits as well, and it rejects the generated harmonics during rectification. But it brings insertion loss and puts a limitation on the bandwidth of the rectifier. For the generated second and third harmonics, the conventional rectifier needs a high performance DC-pass filter as well.

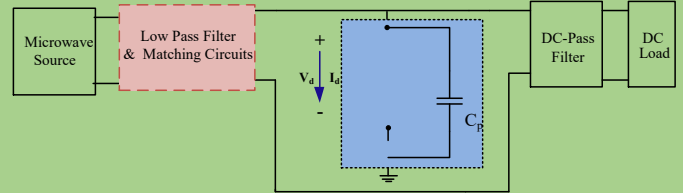


Fig. 1. The conventional microwave with a low-pass filter or a matching circuit.

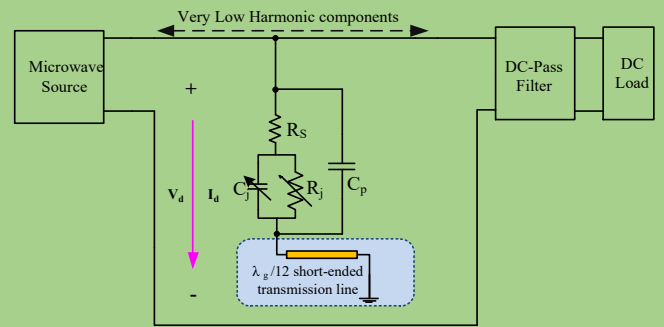


Fig. 2. The proposed rectifier with a third harmonic termination and without the input low-pass filter.

Because the input filter or the matching circuit limits the bandwidth of rectifier, we propose a novel topology which removes the input matching circuit or low-pass filter. Consequently, the operating bandwidth and input power range is expanded in the proposed topology.

The block diagram of the proposed rectifier is shown in

Fig. 2. The $\lambda_g/12$ short-ended transmission line is series connected to the diode. The $\lambda_g/12$ short-end transmission line has following functions.

- i) Presenting high impedance at the third harmonic and relatively high impedance at the second harmonic.
- ii) Compensating the diode capacitive impedance. From the view of impedance matching, the $\lambda_g/12$ short-end transmission line presents $0.58jZ_S$ at fundamental frequency, which compensates diode capacitive reactance where Z_S is the characteristic impedance of the short-ended transmission line.
- iii) Maintaining the DC path for the DC output of the proposed rectifier.

B. Theoretical Analysis

A perfect harmonic termination which presents high impedance at second and third harmonics is hard to realize in practice. Thus, we first focus on the second and third harmonic during rectifying. Considering a rectifier at 5.8 GHz band, the small reactance of C_j at high order harmonic will decrease the conversion efficiency of a Schottky diode. To improve the RF-DC efficiency, we expect to eliminate the high order harmonic current in diode.

As shown in Fig. 2, the diode with a series short-ended transmission line is parallel connected between the microwave source and the DC load. For $\lambda_g/8$ and $\lambda_g/12$, the input impedance of the short-ended transmission line at DC, fundamental frequency, and harmonics are described by (1) and (2), the Z_{SE} is the characteristic impedance of the short-ended transmission line [10].

$$Z_{\lambda_g/8} = jX_L = jZ_{SE} \tan\left(\frac{\pi}{4} \frac{\omega}{\omega_0}\right) = \begin{cases} 0, & \omega = 0 \\ jZ_{SE}, & \omega = \omega_0 \\ \infty, & \omega = 2\omega_0 \\ -jZ_{SE}, & \omega = 3\omega_0 \end{cases} \quad (1)$$

$$Z_{\lambda_g/12} = jX_L = jZ_{SE} \tan\left(\frac{\pi}{6} \frac{\omega}{\omega_0}\right) = \begin{cases} 0, & \omega = 0 \\ 0.58jZ_{SE}, & \omega = \omega_0 \\ 1.73jZ_{SE}, & \omega = 2\omega_0 \\ \infty, & \omega = 3\omega_0 \end{cases} \quad (2)$$

Both the $\lambda_g/8$ and $\lambda_g/12$ presents zero impedance at DC, which maintains the DC path for DC output. The $\lambda_g/8$ has an inductive impedance jZ_{SE} at the fundamental frequency, and turns to an open circuit at the second harmonic $2\omega_0$. By comparison, the $\lambda_g/12$ short-ended transmission line presents inductive $0.58jZ_{SE}$ at the fundamental frequency ω_0 and infinity (open) at the third harmonic $3\omega_0$.

In this work, because the reduced reactance of C_j at relative high frequency, we chose to employ the $\lambda_g/12$ short-ended transmission line to block the third harmonic current inside a HSMS-286B Schottky diode. By referring to paper [11], the Fig. 1 shows the equivalent circuit of the Schottky diode. The diode model consists of a series resistance R_S , a nonlinear junction resistance R_j and a nonlinear capacitance C_j . For a rectifier operating at 5.8 GHz, the C_{j0} of commercial Schottky diode is around 0.1 pF [9]. However, the reactance X_{C_j} will reduce to about 91 Ω at third harmonic (17.4 GHz). Due to the

insufficient reactance of C_j at high order harmonic, the diode can operate at the second harmonic (11.6 GHz) but cannot rectify the third harmonic to DC component. As a result, we focus on suppressing the third harmonic current in a C-band rectifier.

As we can see in (2), the $\lambda_g/12$ short-ended transmission line turns out to be an open circuit at the third harmonic $3\omega_0$, which can attenuates the third harmonic current flows through the diode. Consequently, the power loss on third harmonic is substantially reduced. It turns to inductive impedance of $1.73jZ_{SE}$ at $2\omega_0$, and leads to serious mismatch between the second harmonic impedance and the microwave source. Consequently, inside the diode, the second harmonic can be rectified or mixed with fundamental frequency.

III. RECTIFIER DESIGN

A. Simulation and Comparison

Fig. 3 shows the schematic of propose rectifier. The capacitor C_1 blocks the DC from the rectifier to the microwave source, and the inductor L , C_1 and C_2 form a DC-passed filter. The HSMS-286B diode, capacitor $C_1=3$ pF, $C_2=100$ pF, inductor $L=22$ nH are used.

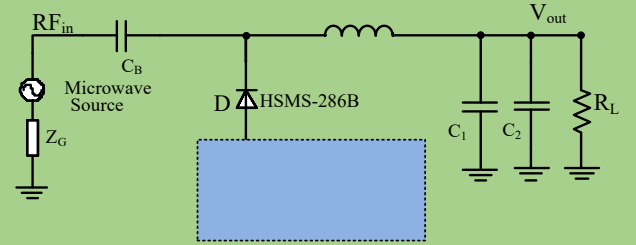


Fig. 3 Schematic of the proposed microwave rectifier, the $\lambda_g/12$ short-ended transmission line is in dash rectangle.

With the harmonic balance simulation, we discuss the reason that why the $\lambda_g/12$ short-ended transmission line can yield the highest RF-DC efficiency. The substrate for the simulation has a thickness of 0.762 mm and dielectric constant $\epsilon_r = 3.65$. With 10 dBm input power and 600 Ω DC load, we have simulated the RF-DC efficiency, harmonic current and harmonic impedance of the proposed rectifier with respect to the length of shorted-ended transmission line from $\lambda_g/12$ to $\lambda_g/7$. In other words, the length of the short-end transmission line varies from 2.58 mm to 4.35 mm, the performance of rectifiers is studied.

In Table I, we present the performance comparison of four rectifiers with different length of the short-ended transmission line. The Z_{in} is the input impedance of the rectifier, and the Z_G is the output impedance of microwave generator. In each case, the width of the short-ended transmission line was adjusted separately to cancel the imaginary part of the rectifier input impedance and then we changed the Z_G of microwave source to match the rectifier.

The input harmonic power level of the four circuits is extracted and each of them is normalized with the input power. For 2.58 mm and 3.04 mm, we can notice that harmonic power at the input port is sufficiently attenuated then the input low-pass filter is not necessary. The highest RF-DC efficiency

of 76.3% is realized at 2.58 mm, and the length turns out to be twelfth-wavelength.

TABLE I

PERFORMANCE COMPARISON WITH FOUR LENGTHS

$Z_G (\Omega)$	50	50	60	60
Length (mm)	2.58	3.04	3.8	4.35
Width (mm)	0.27	0.5	1.1	1.5
$Z_{in} (\Omega)$	$53-j2$	$60-j2$	$68-j5$	$72.8+j0.2$
S_{11} (dB)	-22.4	-20.4	-24.3	-20.5
$R_L (\Omega)$	600	600	600	600
RF-DC Efficiency	76.3%	73.9%	69.5%	65.8%
Input harmonic level at 11.6 GHz (dB)	-19.3	-15.5	-10.1	-7.9
Input harmonic level at 17.4 GHz (dB)	-21.1	-20.4	-20.2	-30.1

Meanwhile, the harmonic impedance of diode is calculated by the ratio between the diode instantaneous voltage and current. The Fig. 4 unfolds clear comparisons between RF-DC efficiency and magnitude of harmonic impedance with regard to the length of the short-ended transmission line. When the length is 2.58 mm, i.e. $\lambda_g/12$, the short-end transmission line presents ultra-high impedance at the third harmonic, which is an open circuit.

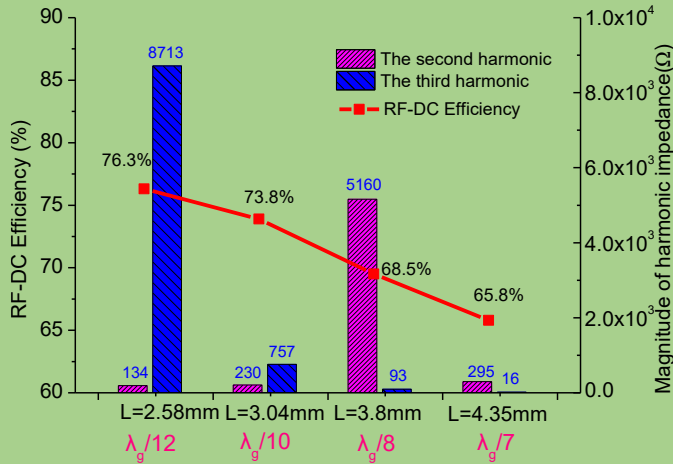


Fig. 4. RF-DC efficiency and the magnitude of the harmonic impedance with respect to the four lengths.

In conclusion, for the highest RF-DC efficiency, the length of the series short-ended transmission line for the proposed topology is twelfth-wavelength and its width is determined by the capacitive impedance of diode and we will discuss it later.

IV. EXPERIMENT VERIFICATION

For validation, the proposed rectifier is fabricated on a 0.76 mm thickness RO-4350B substrate ($\epsilon_r=3.65$) and copper thickness is 35 μm . Fig. 5(a) shows the proposed rectifier with the $\lambda_g/12$ short-ended transmission line and Fig. 5(b) shows the proposed rectifier with the $\lambda_g/8$ short-ended transmission line. The finished rectifier has a very compact dimension of 18 mm by 20 mm.

The simulations results shown below are carried out by Harmonic-Balance method in Keysight ADS. The two kind rectifiers are compared in term of RF-DC efficiency versus operating frequency, input power and DC load. The measurement setup was as follows. We used a signal generator

to power the proposed rectifier. As shown in Fig. 6, a direction coupler was connected between the signal generator and the rectifier. By varying the input power, frequency and R_L , the DC voltage was measured across the load. The power sensor A measures the input power of the rectifying circuit, and the spectrum analyzer measures the reflected fundamental and harmonic power at the input port. The measured RF-DC efficiency was calculated by (3).

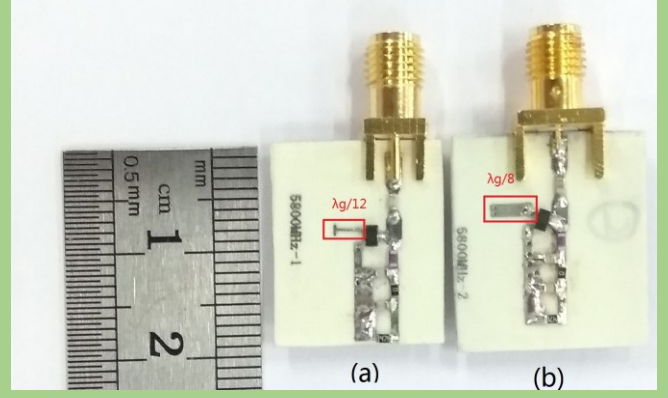


Fig. 5. Photograph of the fabricated microwave rectifier (a) With a $\lambda_g/12$ short-ended transmission line. (b) With a $\lambda_g/8$ short-ended transmission line.

$$\eta_{RF-DC} = \frac{P_{DC}}{P_{IN}} \times 100\% = \frac{V_o^2}{R_L} \times \frac{1}{P_{IN}} \times 100\% \quad (3)$$

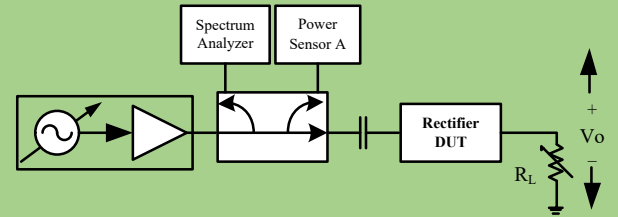


Fig. 6 Experiment architecture of the RF-DC efficiency and the reflected harmonic.

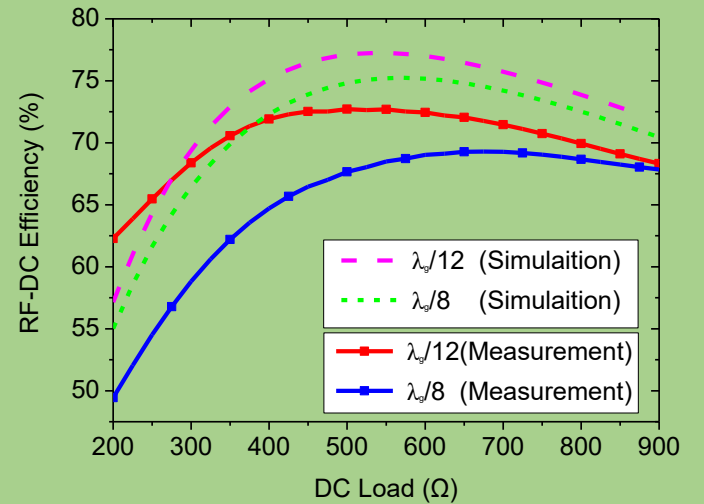


Fig. 7. Comparison of the RF-DC efficiency versus output load for an input power of 10 dBm at 5.8 GHz.

The simulation and measurement efficiencies versus output

DC load are presented in Fig. 7. The rectifier with the $\lambda_g/12$ maintains high efficiency for a wider output load range than the one with the $\lambda_g/8$. With the $\lambda_g/12$, a peak measured RF-DC efficiency of 72.7% occurs with 500 Ω DC load, and efficiency maintains above 70% when DC load varies from 350 Ω to 800 Ω . The measured results correspond with the simulated results.

The simulation and measurement RF-DC conversion efficiencies of the rectifiers with the $\lambda_g/12$ and with the $\lambda_g/8$ at 10 dBm are depicted at Fig. 8. It can be seen that greater than 60% measurement efficiency of the rectifier with the $\lambda_g/12$ can be obtained from 5.25 GHz to 6.35 GHz while that of the rectifier with the $\lambda_g/8$ is from 5.27 GHz to 6.07 GHz. By comparison, the rectifier with the $\lambda_g/12$ shows wider bandwidth that the RF-DC efficiency maintains more than 70% from 5.55 GHz to 6.2 GHz. There exists slight difference between the simulated and measured results, which is maybe caused the self-resonance of the lumped DC-pass filter.

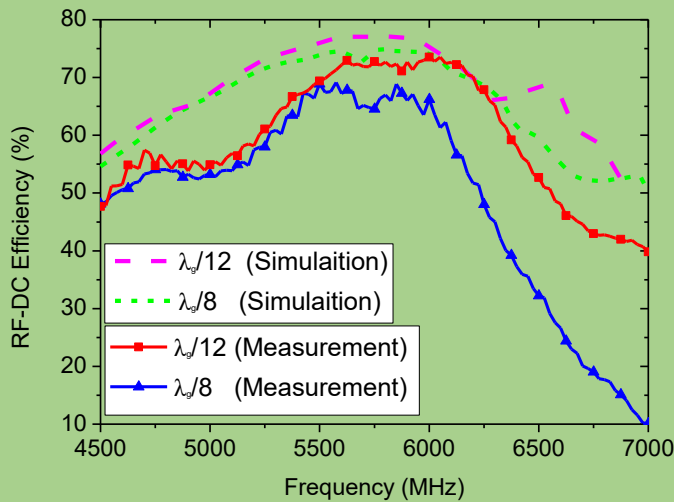


Fig. 8. Comparison of the RF-DC efficiency versus operating frequency with $R_L = 500 \Omega$ at 10 dBm.

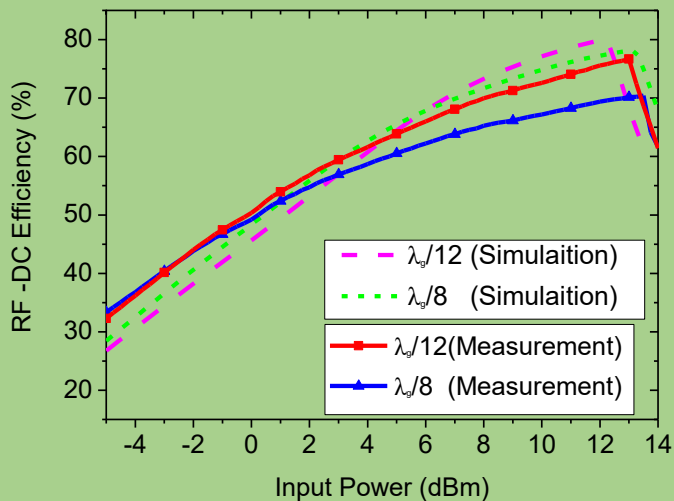


Fig. 9. Comparison of the RF-DC efficiency versus input power with 500 Ω DC load at 5.8 GHz.

Fig. 9 shows the simulation and measurement efficiencies versus input power. With the $\lambda_g/12$, the peak measured

efficiency of the proposed rectifier is 76.6% at 13 dBm. It can be observed that the measured efficiency maintains more than 70% from 8.25 dBm to 13.25 dBm, and better than 60% from 3.5 dBm to 14 dBm. The input power range corresponding to high efficiency of the rectifier with the $\lambda_g/12$ is wider than that of the one with the $\lambda_g/8$. Such comparison shows that the RF-DC efficiency is almost the same at low incident power and almost 6 percent higher at the maximum in the $\lambda_g/12$ type than in the $\lambda_g/8$ type. The measured results agree well with the simulated results. There is still some slight difference between the measurement and simulation at low input power, and the uncertainty of diode model as well as the tolerance of PCB fabrication is the reason for the discrepancy.

V. CONCLUSION

This paper has presented a novel method for designing high efficiency rectifier with wide input power range and frequency bandwidth. This design is based on the principle of harmonic termination and elimination of the input low-pass filter. As the input cascading band-pass or low-pass filters in conventional microwave rectifiers has been removed, the bandwidth and input power range of the proposed rectifier is extended remarkably.

For the proposed rectifier with the $\lambda_g/12$ transmission line at 5.8 GHz, the input power range for RF-DC efficiency over 70% is 5 dB, a wide input power range of 10.5 dB for efficiency over 60% can be obtained. Meanwhile, the fractional bandwidth corresponding to over 70% efficiency is 11.9%.

REFERENCES

- [1] N. Shinohara, "Power without wires," *IEEE Microw. Mag.*, vol. 12, no. 7, pp. S64–S73, Dec. 2011.
- [2] W. Huang, B. Zhang, X. Chen, K.-M. Huang, and C.-J. Liu, "STUDY ON AN S-BAND RECTENNA ARRAY FOR WIRELESS MICROWAVE POWER TRANSMISSION," *Prog. Electromagn. Res.*, vol. 135, pp. 747–758, 2013.
- [3] Y. H. Kim, T. H. Ki, C. Chung, Y. K. Moon, Y. Lim, and S. o Lim, "Implementation of a low-cost and low-power batteryless transceiver SoC for UHF RFID and wireless power transfer system," in *2012 42nd European Microwave Conference*, 2012, pp. 514–517.
- [4] T. Paing, E. Falkenstein, R. Zane, and Z. Popovic, "Custom IC for Ultra-low Power RF Energy Harvesting," in *2009 Twenty-Fourth Annual IEEE Applied Power Electronics Conference and Exposition*, 2009, pp. 1239–1245.
- [5] Y.-H. Suh and K. Chang, "A high-efficiency dual-frequency rectenna for 2.45- and 5.8-GHz wireless power transmission," *IEEE Trans. Microw. Theory Tech.*, vol. 50, no. 7, pp. 1784–1789, Jul. 2002.
- [6] H. Sakaki and K. Nishikawa, "Broadband rectifier design based on quality factor of input matching circuit," in *2014 Asia-Pacific Microwave Conference*, 2014, pp. 1205–1207.
- [7] X. Y. Zhang, Z. X. Du, and Q. Xue, "High-Efficiency Broadband Rectifier With Wide Ranges of Input Power and Output Load Based on Branch-Line Coupler," *IEEE Trans. Circuits Syst. Regul. Pap.*, vol. PP, no. 99, pp. 1–9, 2016.
- [8] D. Wang, X. A. Nghiem, and R. Negra, "Design of a 57% bandwidth microwave rectifier for powering application," in *2014 IEEE Wireless Power Transfer Conference*, 2014, pp. 68–71.
- [9] T. W. Yoo and K. Chang, "Theoretical and experimental development of 10 and 35 GHz rectennas," *IEEE Trans. Microw. Theory Tech.*, vol. 40, no. 6, pp. 1259–1266, Jun. 1992.
- [10] D. M. Pozar, *Microwave engineering*. John Wiley & Sons, 2009.
- [11] J. O. McSpadden, L. Fan, and K. Chang, "Design and experiments of a high-conversion-efficiency 5.8-GHz rectenna," *IEEE Trans. Microw. Theory Tech.*, vol. 46, no. 12, pp. 2053–2060, Dec. 1998.